

CS348 Computer Networks MidSem, Sat 12 Sep 2026, 8-10am, Max. Marks: 30

Name: _____ Roll number: House: _____

General instructions

- Write only in the given space. Answer briefly but crisply. **Strong recommendation: use DARK PENCIL**
- You are allowed to refer to your own hand-written notes only. You do NOT need a calculator.
- All answers have to be (briefly) explained. State any necessary assumptions. No clarifications during exam.
- The final answer of each question has to be entered in the given box.

Q1: [2 x 5 = 10 marks] Short answer questions

1. I once bought a wired earphone to use while watching videos. I found that if I played at twice the video speed, the video was fine but the audio was choppy/unclear only when I used the earphone (not when using the laptop speaker). What might be the reason behind this? Explain *using concepts learnt in this course*.

Reason: _____

2. Ms. H. D. Elsi is experimenting with a new flag 01010101 to use as the start/end sentinel in framing. In order to prevent data bits from being confused with the flag, she decided to stuff a one after receiving the data sequence 010101. Will this work? Justify.

Answer (Y/N):

Reason: _____

3. Suppose that a communication link has 4 MHz bandwidth. What is the achievable capacity of this channel under any encoding, if the SNR is 36dB? (Make suitable engineering approximations; 3dB = factor of 2)

Answer (closest integer, in Mbps):

Calculation: _____

4. Explain the following *using concepts learnt in this course*. (a) use of a flyover to reduce congestion at a road intersection, (b) the intersection at IIT main gate switches to flashing yellow lights during late night

(a): _____

(b): _____

5. Among the functionalities of (i) medium access control (MAC), (ii) retransmission, (iii) forward-error-correction, (iv) error-detection, which are provided by:

(a) WiFi: _____

(b) Fully Switched Ethernet: _____

Q2: IP fragmentation and pipelining [3+1=4 marks]: In the network below, N1 sends a packet of size P bytes (IP payload) to N2 via IP routers R1 and R2. Assume that TCP and IP headers are both 20 bytes each. Ignore propagation and processing delays, and assume that there is no other traffic in the network. Consider two scenarios: S1 where the packet is sent as a single IP fragment, and S2 where the packet is split into 3 fragments of equal size.



1. For what values of P will scenario S1 have a larger delay for the packet, compared to scenario S2 ?

Answer:

Reason:

2. What is the minimum value of P for which scenario S2 is possible?

Answer:

Calculation:

Q3: Forwarding [1x4 = 4 marks]: The forwarding table at a router in the global Internet is as shown below. The prefixes/addresses are represented in hexadecimal for ease of conversion to binary.

Network Prefix	Next-hop
C4.5E.2.0/23	A
C4.5E.4.0/22	B
C4.5E.C0.0/19	C
C4.5E.40.0/18	D
C4.4C.0.0/14	E
C0.0.0.0/2	F
80.0.0.0/1	G

1. Which next hop will a packet with destination address C4.4B.31.2E be routed to?

Answer:

Calculation:

2. Which next hop will a packet with destination address C4.5E.05.09 be routed to?

Answer:

Calculation:

3. Which next hop with a packet with destination address C4.4D.31.2E be routed to?

Answer:

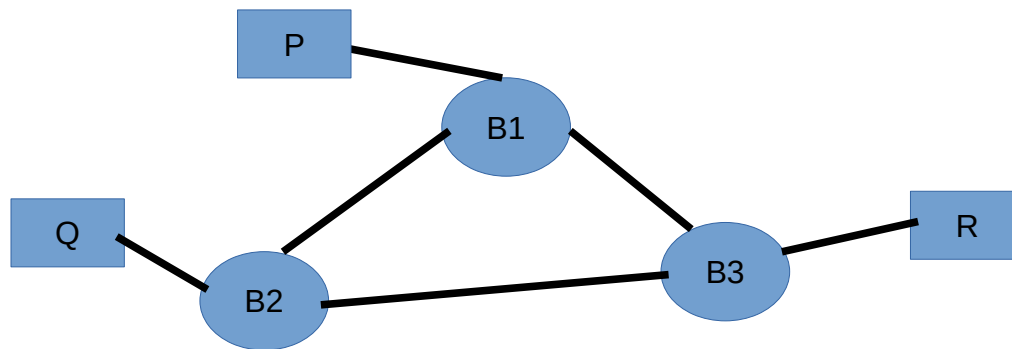
Calculation: _____

4. Which next hop with a packet with destination address C0.A8.1.2 be routed to?

Answer:

Calculation: _____

Q4: Ethernet [2+3+2=7 marks]: In the switched Ethernet below, B1, B2, B3 are bridges; P, Q, R are end-hosts.



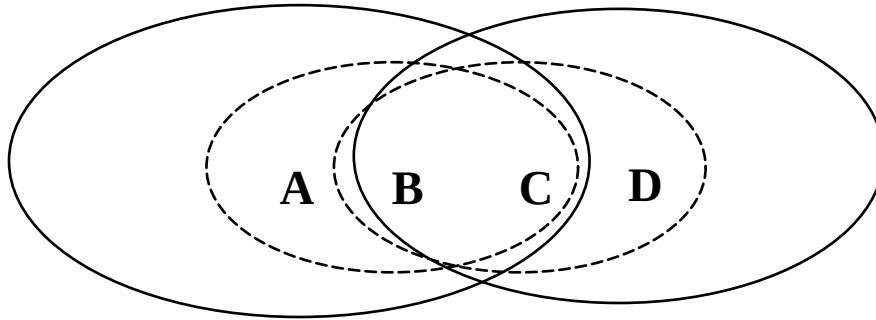
1. Suppose the spanning tree protocol has stabilized. In the above diagram itself, mark each port of each bridge as RP (root port), DP (designated port), or IN (inactive).
2. Suppose that IN (inactive port) means that no packet will be sent or received on that port. Construct a short sequence of events E1, E2, E3... (as many as needed) to show what *undesirable* thing can happen with such a definition of IN.

Answer: _____

3. Given the above, what exactly *does* marking a port as IN (inactive) mean? What exactly can or cannot be sent/received on such a port?

Answer: _____

Q5: Wireless and MAC [0.5+0.5+1+1+2=5 marks]: Consider the wireless network below. The transmit ranges of each node is as shown. A can be heard by B and C but not by D. B can be heard by A and C but not by D. C can be heard by B and D but not by A. D can be heard by B and C but not by A. Suppose A is sending data to B and C is sending data to D. Both are sending packets as fast as is allowed by the MAC. There is no other traffic. Ignore header overheads and propagation and processing delays.



1. Mention one hidden node scenario and one exposed node scenario in the above network.

Hidden node scenario: _____

Exposed node scenario: _____

2. Suppose no MAC is employed i.e. there is nothing to detect or avoid collisions. There are no ACKs either. What is the received throughput at B as a percentage of A's sending rate?

Answer (percentage, nearest integer):

Reason: _____

3. Now suppose that each node uses CSMA/CA with a fixed contention window. There is no RTS/CTS or link layer ACK/retransmission. What is the received throughput at B as a percentage of A's sending rate?

Answer (percentage, nearest integer):

Reason: _____

4. In the same situation as above, what is the received throughput at D as a percentage of C's sending rate?

Answer (percentage, nearest integer):

Reason: _____

5. Now consider that each node uses CSMA/CA with a fixed contention window. There is no RTS/CTS, but there is WiFi-like immediate ACK and up to 1 retry (so 2 tries including original transmission). What is the received throughput at D as a percentage of the link speed?

Answer (percentage, nearest integer):

Reason: _____
